

Research Paper 4 – Optimizing Delegation Between Human and AI

1. Core Problem

In a hybrid team of humans and AI agents, neither the human nor the AI is always the best performer for a given task. This paper proposes a "manager agent" that learns an optimal delegation policy to decide who (human or AI) should perform an action at any given time to maximize overall team performance and reduce costs.

2. Key Innovation: Mismatched Models

Unlike previous work that assumes humans and AI operate under the same model of the world (a single Markov Decision Process or MDP), this framework allows them to have different, compatible models.

- **Shared State Space:** Both agents understand the same set of possible states.
- **Different Actions & Transitions:** The human and AI can have different available actions and different ways of moving between states. For example, an AI might take a direct path, while a human might take a more scenic but longer route.
- **Compatibility Requirement:** As long as a state reachable by one agent is also eventually reachable by the other, their models are considered compatible.

3. The Manager Agent's Model

The manager agent learns its policy by observing the outcomes of its delegations, without needing to know the internal logic or private knowledge of the human or AI workers.

- **Managed MDP (MMDP):** The manager views the team's operation as a combined MDP where its "actions" are delegating to either the human or the AI.
- **Learning Through Observation:** The manager learns which agent is likely to perform better in a given state by associating its delegation choices with the resulting outcomes (success, failure, cost).
- **Inspired by SMDPs:** The model is inspired by Semi-Markov Decision Processes (SMDPs), especially the "options" framework. Delegating to an agent is treated like selecting an "option," where the agent might perform a sequence of actions before returning control.

4. Experimental Setup

The approach was tested in a "gridworld" environment (a maze-like grid) where agents had to navigate from a start point to a goal.

- **Agent Variation:** Agents differed in their "step size" (how many squares they could move at once) and their "error likelihood" (how often they made a mistake).
- **Cost Factoring:** The manager's reward function was designed to penalize agent costs and the total time taken to reach the goal. This forced the manager to learn a trade-off between speed, cost, and risk.

5. Results

The manager agent successfully learned to delegate effectively and consistently outperformed a "random manager" that delegated with a 50/50 chance.

- **Adaptation to Cost:** The manager adapted its strategy based on the costs assigned to different agents. For example, if short-step agents were cheaper, it used them more, and vice versa.
- **Performance Under Uncertainty:** Even when one agent in the team was highly error-prone, the manager learned to rely on the more stable agent, still achieving high scores.

6. Conclusion

The paper successfully demonstrates a method to train a manager agent that can effectively orchestrate a hybrid human-AI team, even when the team members have fundamentally different ways of operating. The key is that the manager can learn a successful delegation policy purely by observing outcomes, without needing access to the internal models of the agents it manages.